



Faculty of Computer Science

Master of Data Science

Applied Quantitative Logistics

Genetic Speciation Algorithm for Addressing Combinatorial Optimization Problems: Application to the Vehicle Routing Problem



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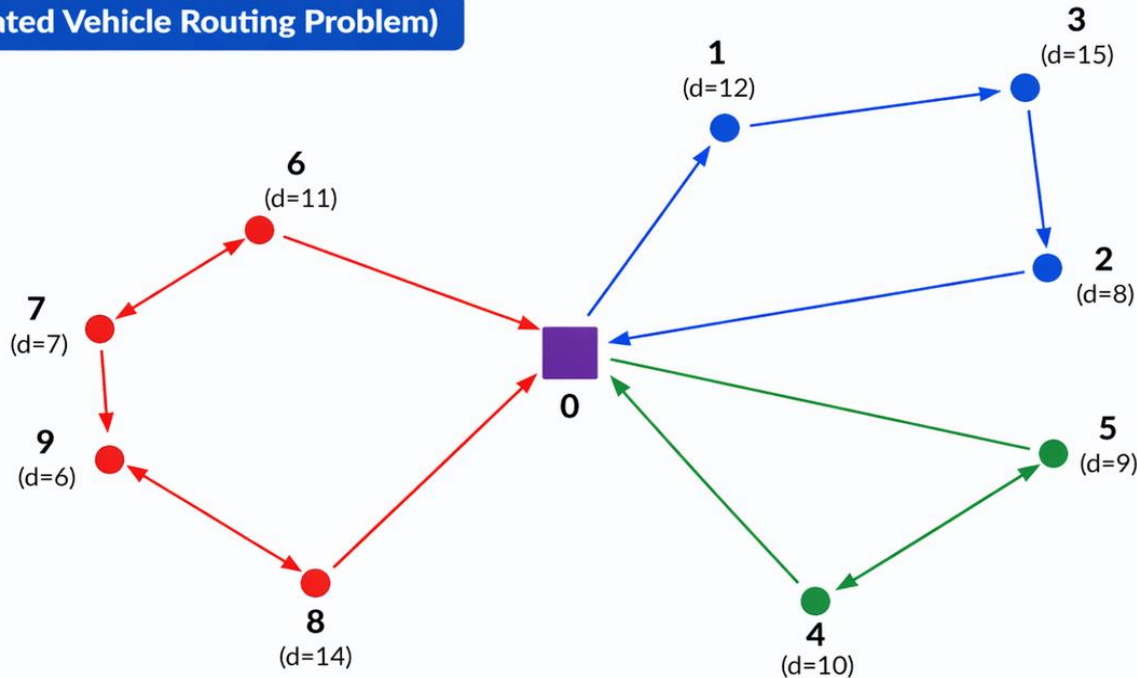
Problem Definition

What is CVRP?

- A fleet of vehicles departs from a single depot to serve a set of customers
 - Each customer has a fixed demand; each vehicle has a capacity limit
 - Objective: minimize total travel distance across all routes
- NP-hard combinatorial optimization problem
 - Search space grows factorially with customer count

Why It's Hard?

CVRP (Capacitated Vehicle Routing Problem)



Route 1
0 → 1 → 3 → 2 → 0 Distance = 31

Route 2
0 → 5 → 4 → 0 Distance = 18

Route 3
0 → 6 → 7 → 9 → 8 → 0 Distance = 29

Total Distance = 31 + 18 + 29 = 78

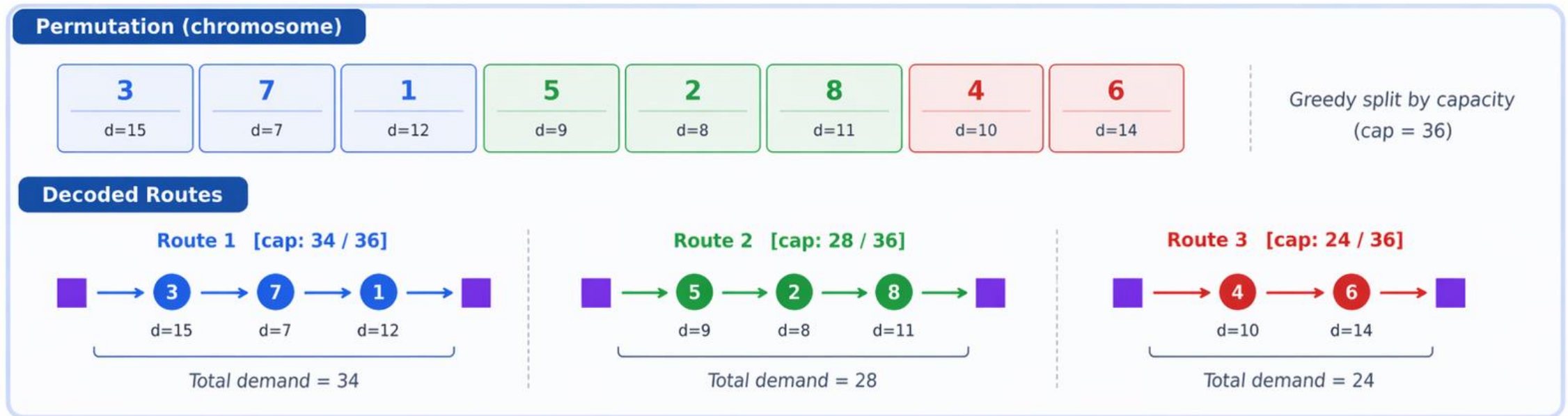
Solution Representation

Solution Representation

- A solution is encoded as a permutation of all customers
- Routes are decoded greedily: a new route begins whenever adding the next customer would exceed vehicle capacity
- Fitness = total route length including depot returns

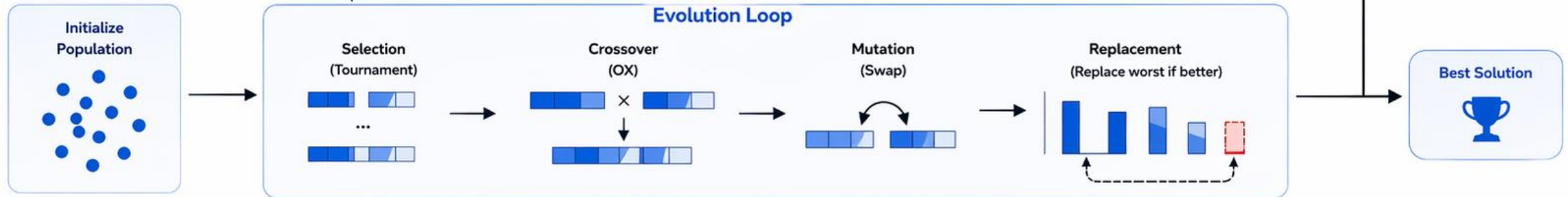
Fitness Evaluation

- Single-pass over the permutation — no $O(n^2)$ feasibility checks
- Most frequently called function in the entire pipeline; kept minimal for performance



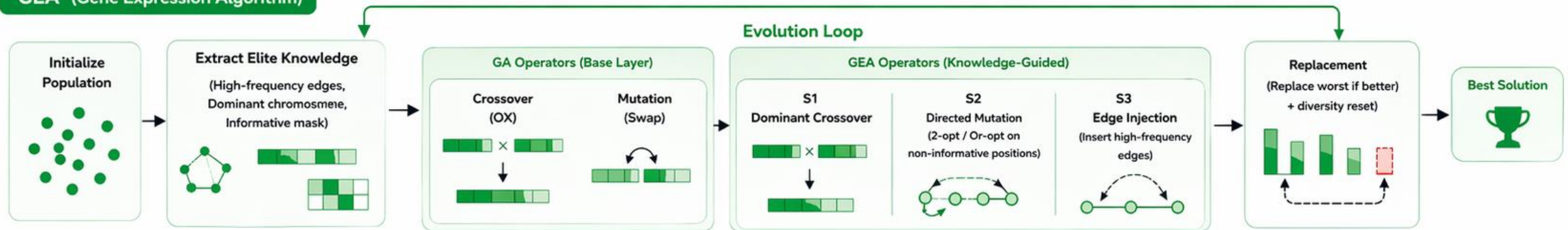
GA (Genetic Algorithm)

Repeat until maximum generations



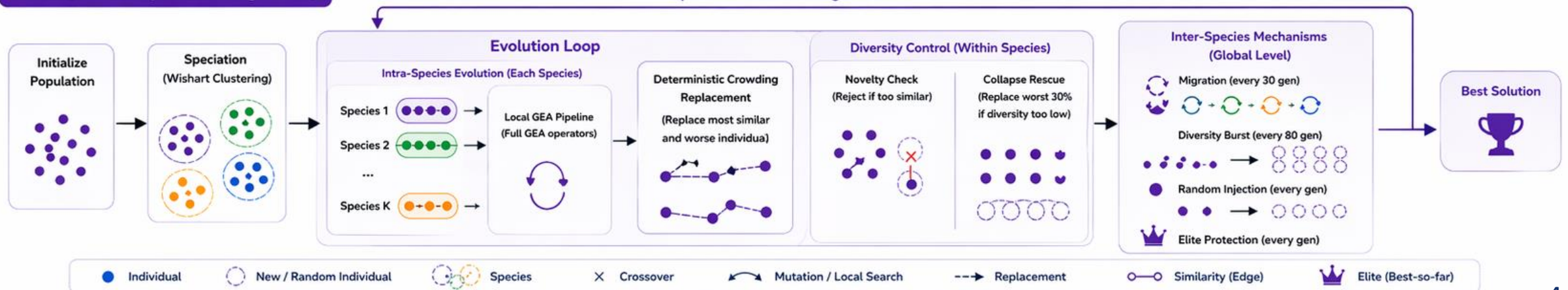
GEA (Gene Expression Algorithm)

Repeat until maximum generations



GSA (Genetic Speciation Algorithm)

Repeat until maximum generations



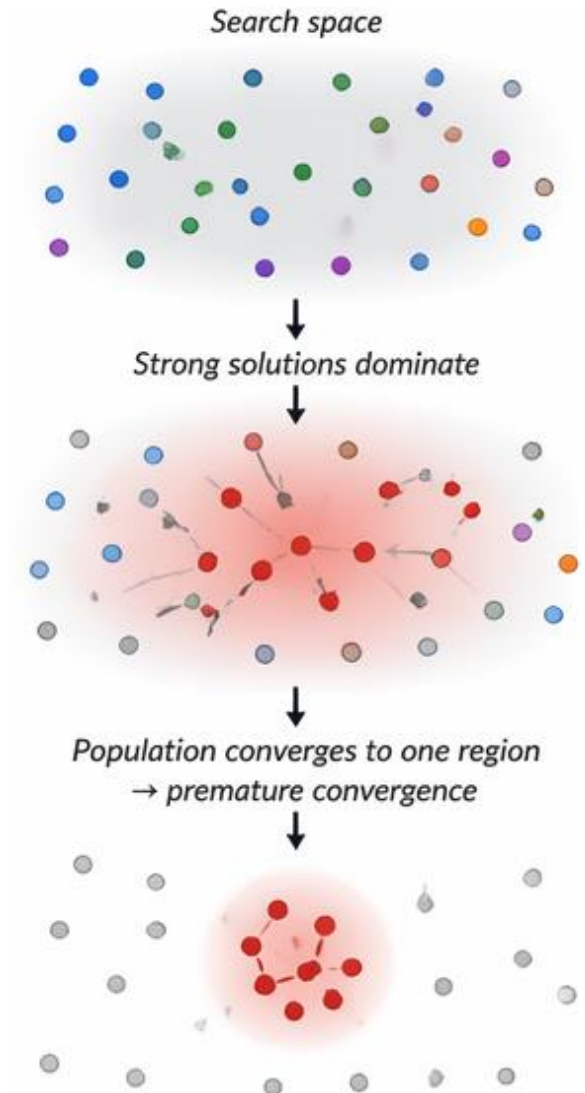
Wishart Speciation

Why Speciation?

- Single-population methods force all individuals to compete for the same slots
- Strong solutions dominate → structurally different individuals are eliminated
- The population converges to one region of the search space → premature convergence
- Speciation enforces structural separation: diverse archetypes coexist and evolve in parallel

How to measure similarity between individuals?

- Primary (80%): Jaccard distance between edge sets — two individuals are close if they share many consecutive customer pairs
- Secondary (20%): fitness rank distance — prevents grouping individuals of very different quality together
- Combined: $D = 0.8 \times \text{Jaccard} + 0.2 \times \text{rank distance}$



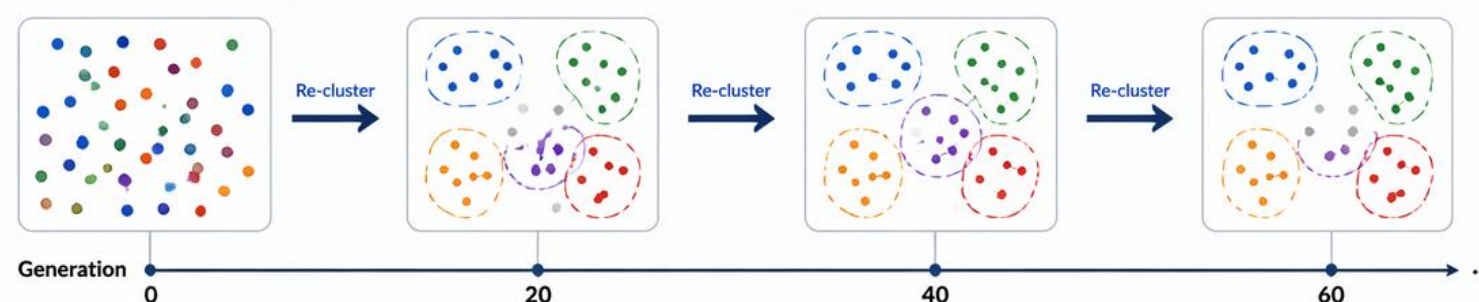
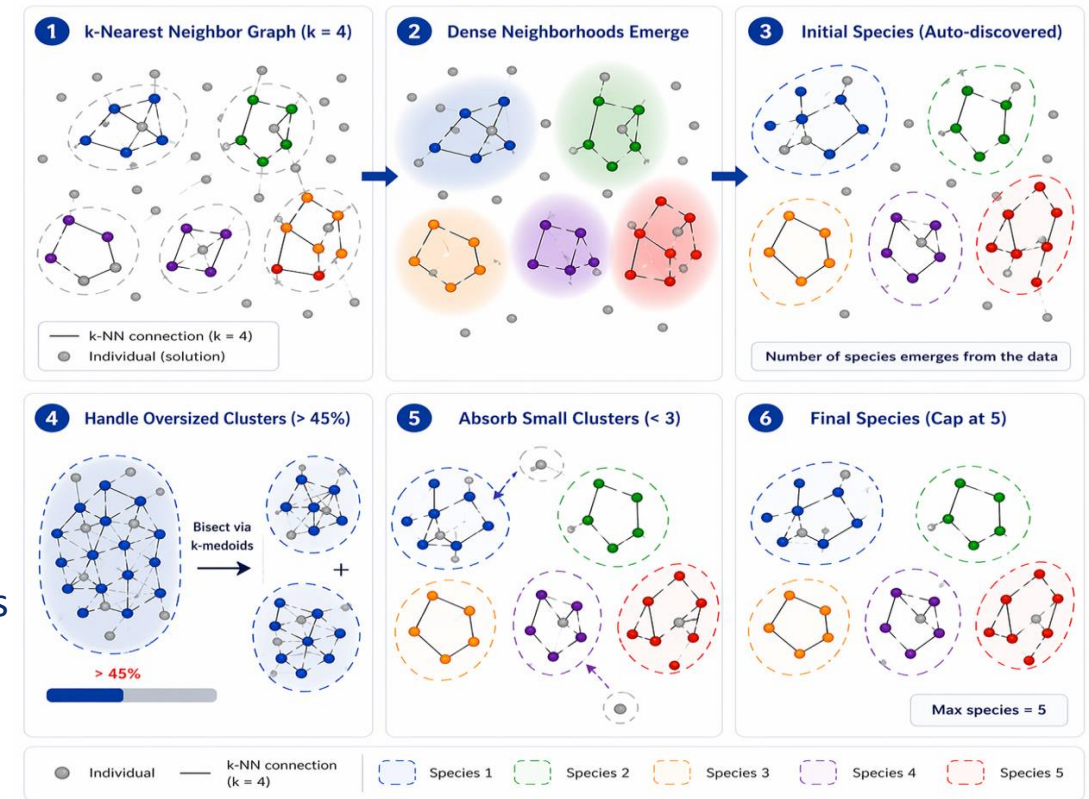
Wishart Speciation

How species are formed?

- Connect each individual to its k nearest neighbors ($k = 4$)
- Clusters form around dense neighborhoods and expand naturally
- Number of species emerges from the data — not fixed in advance
- Oversized clusters ($> 45\%$ of population) are bisected via k -medoids
- Clusters smaller than 3 individuals are absorbed into the nearest species
- Maximum species count capped at 5 to prevent over-fragmentation

When to Re-cluster?

- Every 20 generations: routine re-clustering to reflect evolved population structure
- Immediately after a diversity burst: population has changed drastically, old species labels are invalid



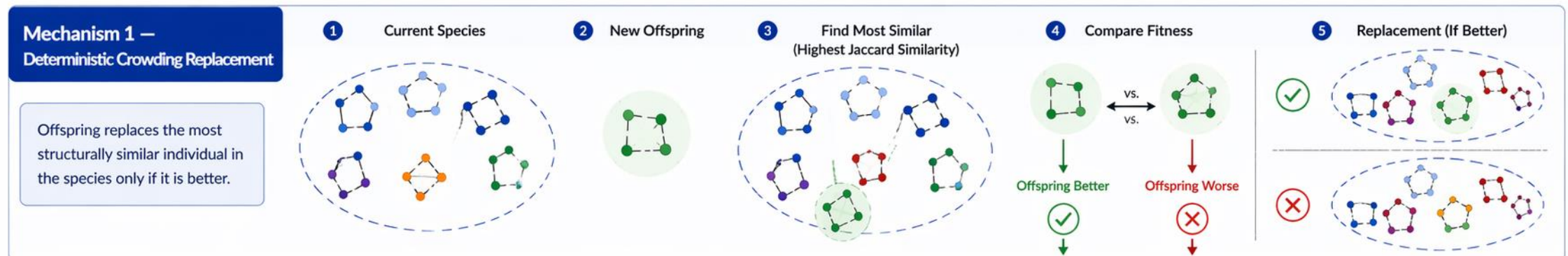
Intra-Species Diversity Control

Why Not Just Replace the Worst?

- GEA uses replace-worst: an offspring replaces the globally worst individual if it is better
- This works in a single population but breaks down inside a species — the worst individual may be structurally very different from the offspring, so replacing it destroys diversity rather than improving it
- GSA uses three targeted mechanisms instead

Mechanism 1 — Deterministic Crowding Replacement

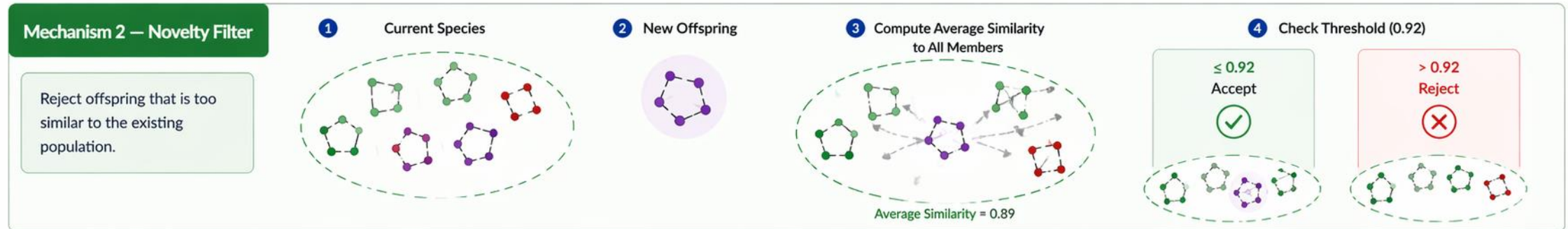
- When an offspring is produced, find the most structurally similar individual in the species (highest Jaccard similarity)
- The offspring replaces that individual only if it has better fitness
- Effect: new solutions displace their structural competitors, not unrelated individuals — the species retains its structural spread



Intra-Species Diversity Control

Mechanism 2 — Novelty Filter

- Before accepting an offspring into the species, check its average similarity to all current members
- If the offspring is too similar to the existing population (similarity > 0.92), reject it outright
- Effect: prevents the species from collapsing to a single structure even under strong selective pressure



Mechanism 3 — Collapse Rescue

- Every generation, compute the average pairwise Jaccard distance within each species
- If this falls below 0.04 (near-identical structures across all members), trigger a rescue
- Replace the worst 30% of the species with randomly generated individuals
- Effect: breaks up clone clusters before they waste further generations going nowhere

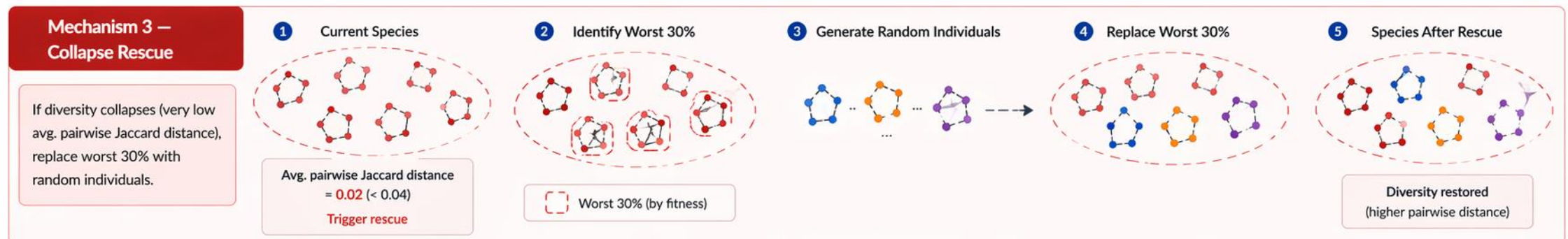
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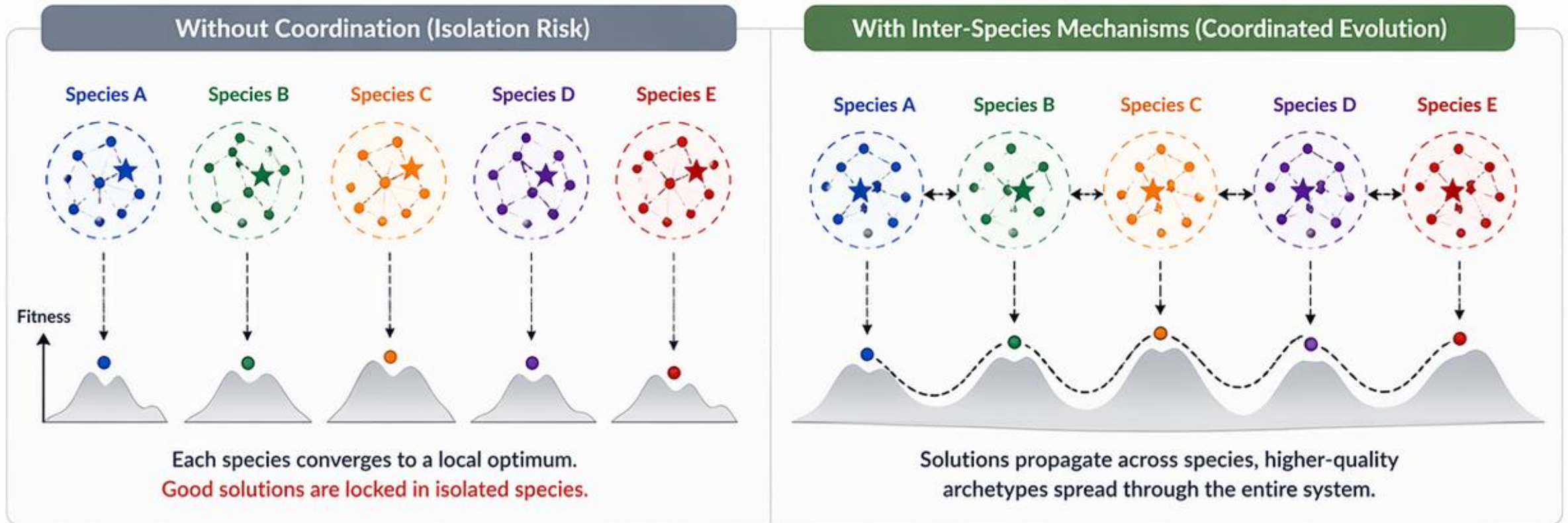
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Inter-Species Mechanisms

Why Inter-Species Mechanisms?

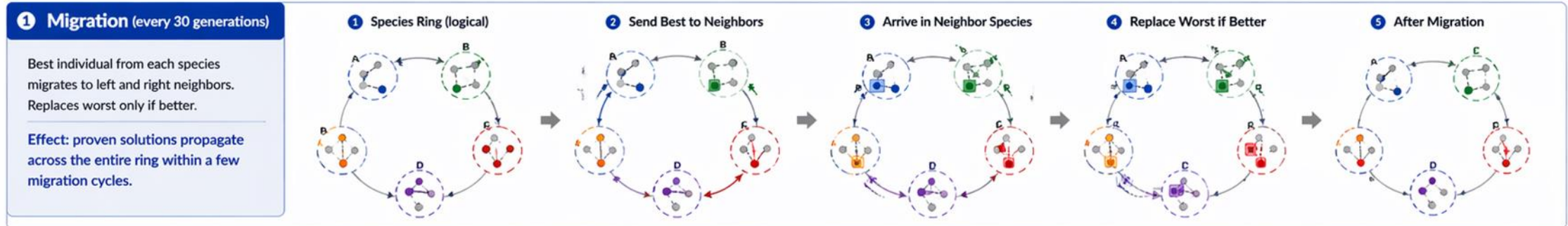
- Independent species evolution risks isolation — a species may converge to a locally strong but globally inferior solution archetype
- Three periodic mechanisms coordinate across species to share discoveries and prevent system-wide stagnation



Inter-Species Mechanisms

Migration (every 30 generations)

- Species are arranged in a logical ring
- The best individual from each species is sent to both its left and right neighbors
- In each receiving species, the migrant replaces the worst member only if it has better fitness
- Effect: proven solutions propagate across the entire ring within a few migration cycles — good structural patterns are not locked inside a single species



Diversity Burst (every 80 generations)

- The bottom 25% of the global population is replaced with randomly generated individuals
- Forces a re-clustering on the next cycle, introducing entirely new structural material
- Effect: resets stagnating species before they waste too many generations; the long interval (80 gens) avoids disrupting ongoing convergence

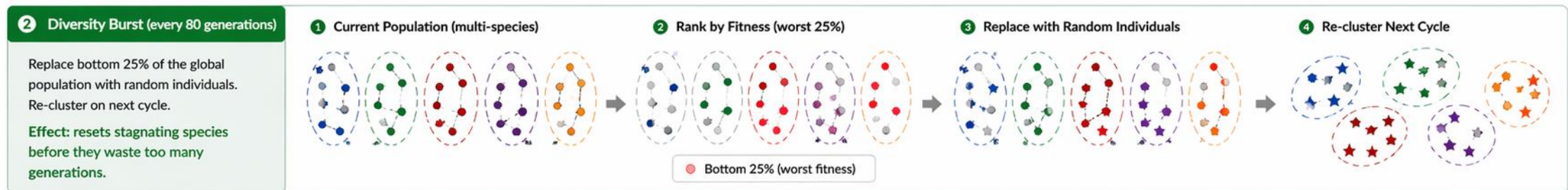
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Inter-Species Mechanisms

Random Injection (every generation)

- The worst 3% of the global population is replaced with random individuals, but only if the random individual is actually better
- A lightweight continuous mechanism providing a steady trickle of new genetic material
Effect: prevents full stagnation without disturbing the overall population structure



Elite Protection (every generation)

- The global best solution found so far is stored at the start of each generation
- After all operators have run, if the global best has been accidentally overwritten, it is restored
- Effect: guarantees monotonic improvement in reported best fitness — no good solution is ever permanently lost



Experimental Setup

Benchmark Instances

Instance	Customers	Vehicles
A-n32-k5	31	5
A-n60-k9	59	9
A-n80-k10	79	10

Algorithms Compared

- GA: crossover + mutation only (baseline)
- GEA: GA + elite knowledge extraction + S1/S2/S3 operators
- GSA: GEA + Wishart speciation + intra-species diversity control + inter-species mechanisms

Evaluation Metrics

- Best: best solution found across all 5 runs
- Mean: average solution quality across runs
- Std: standard deviation across runs (consistency)
- Avg Time: average wall-clock time per run in seconds

Hyperparameters

Parameter	Value
Population size	200
Generations	500
Independent runs	5
Base random seed	42
Elite ratio	20% (GEA) / 40% (GSA)
Max species	5
Re-cluster every	20 generations

Reproducibility

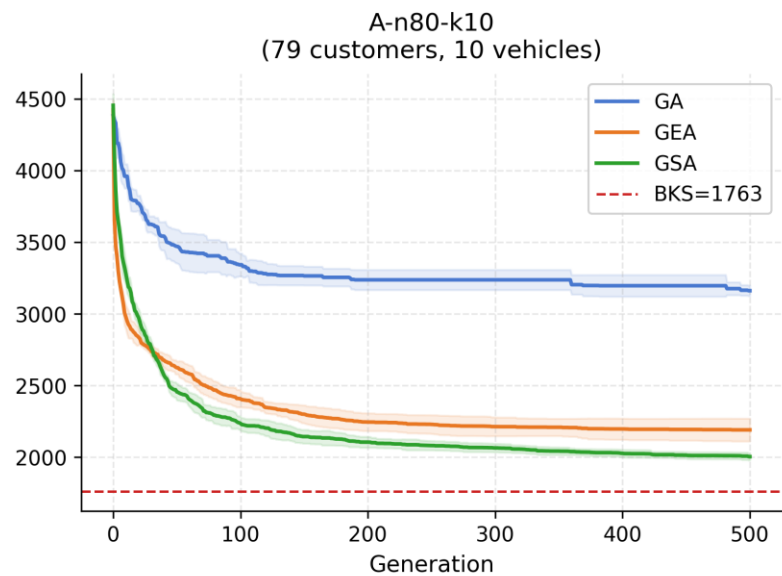
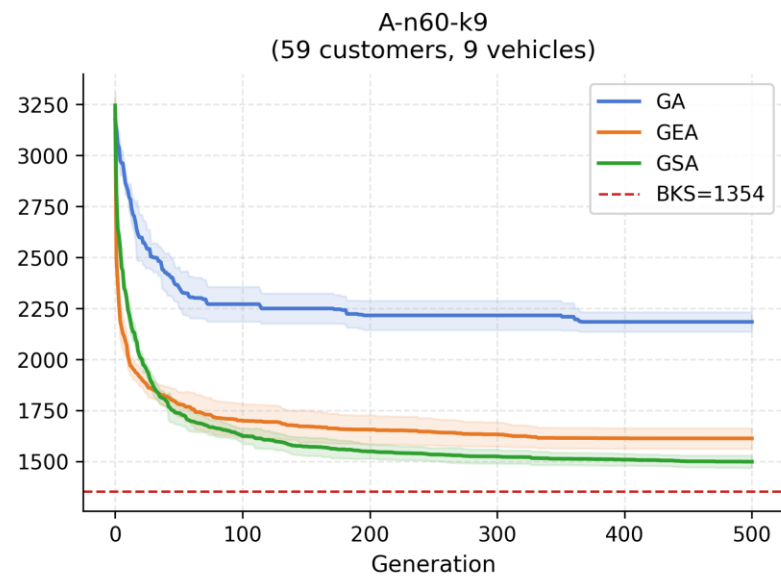
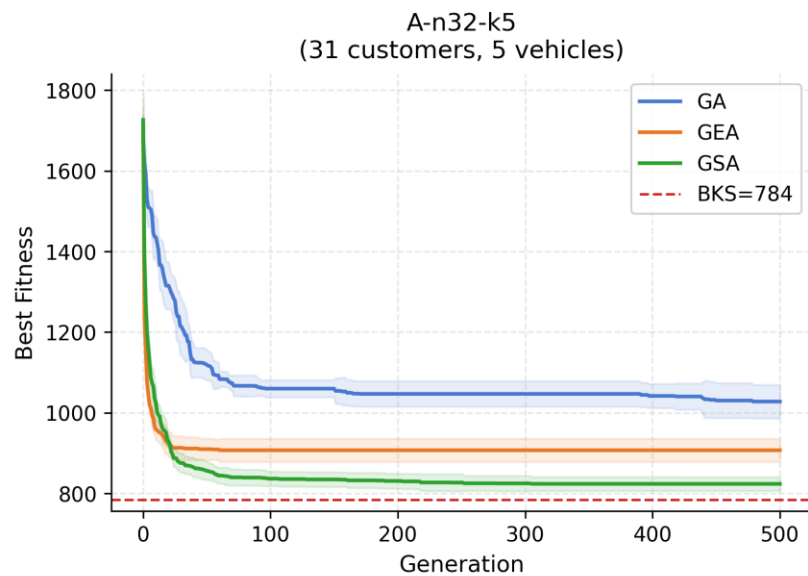
- Each algorithm and run uses a distinct random seed derived from the base seed, instance name, and run index
- Every reported solution is validated before recording: permutation must be complete and every route must satisfy capacity constraints



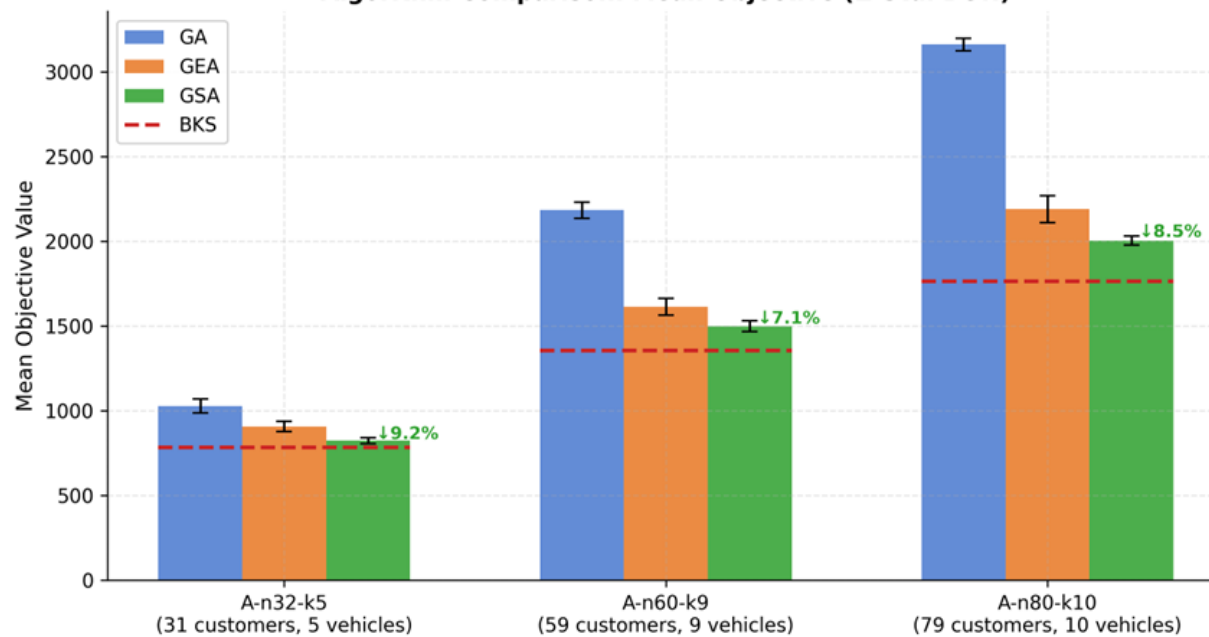
Experimental Results

Instance	Algorithm	Best	Mean	Std. Dev.	Gap to BKS	Avg. Time
A-n32-k5	GA	944.6	1027.9	41.79	31.1%	4.5s
A-n32-k5	GEA	857.6	907.1	28.99	15.7%	12.1s
A-n32-k5	GSA	803.3	823.6	17.27	5.1%	55.5s
A-n60-k9	GA	2110.1	2184.1	48.12	61.3%	6.8s
A-n60-k9	GEA	1517.2	1613.0	50.63	19.1%	17.5s
A-n60-k9	GSA	1463.7	1499.0	30.72	10.7%	90.9s
A-n80-k10	GA	3104.7	3161.3	36.98	79.3%	8.0s
A-n80-k10	GEA	2078.1	2190.9	79.58	24.3%	20.7s
A-n80-k10	GSA	1978.1	2004.2	25.30	13.7%	132.4s

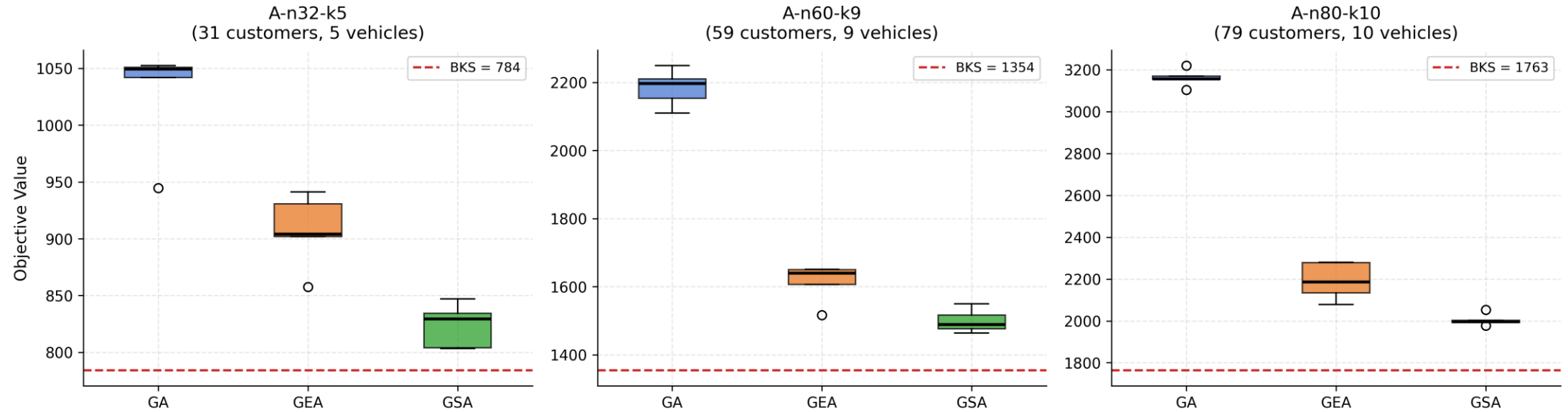
Mean Convergence Curves (± 1 Std. Dev.)



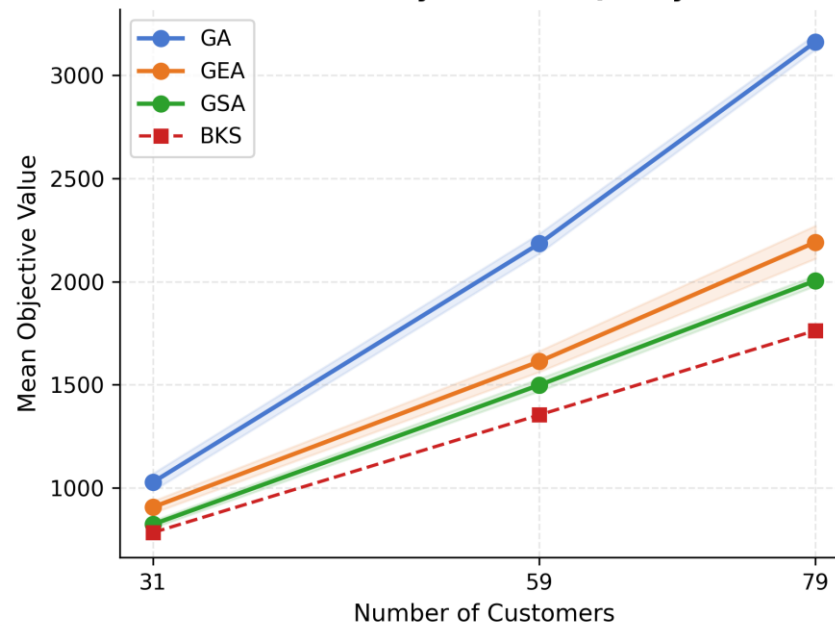
Algorithm Comparison: Mean Objective ($\pm$ Std. Dev.)



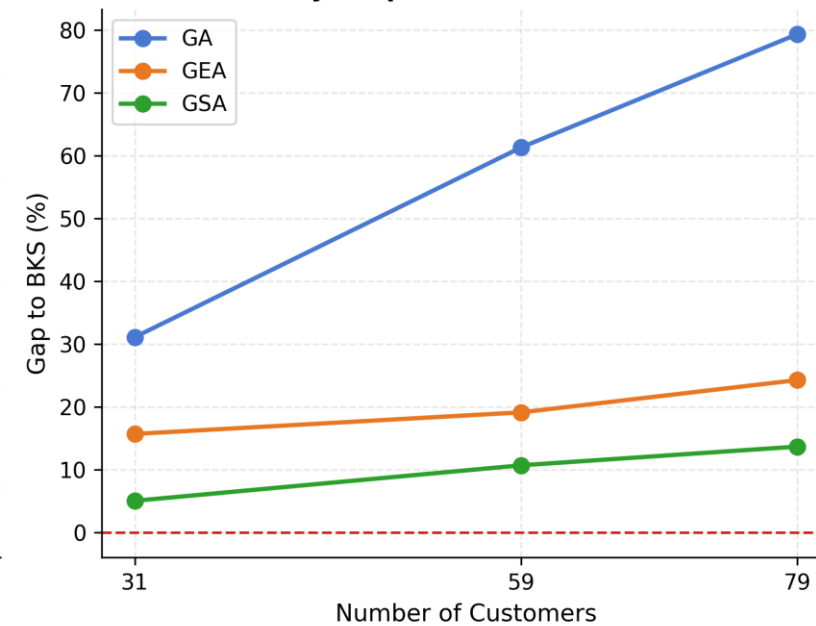
Result Distribution Across Independent Runs



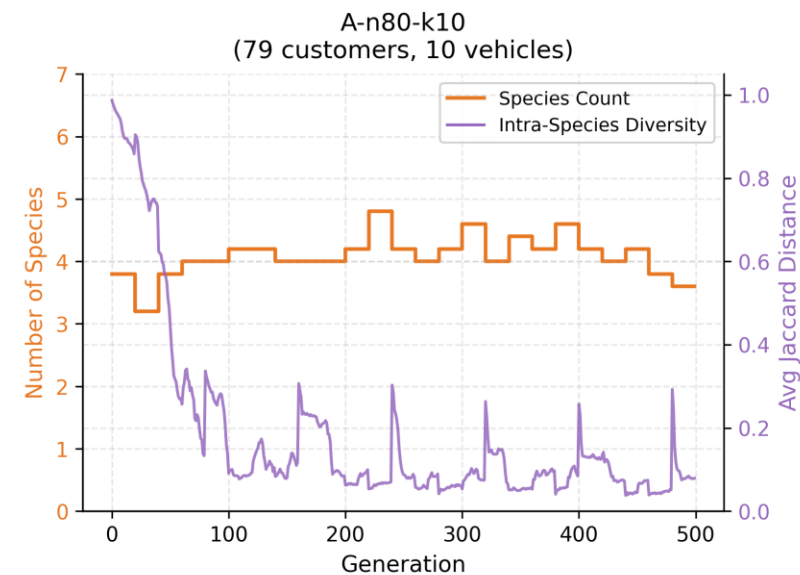
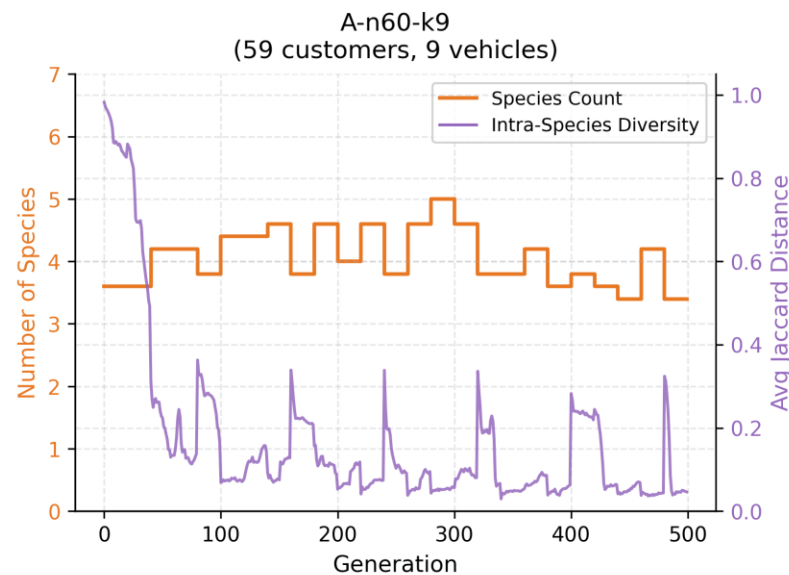
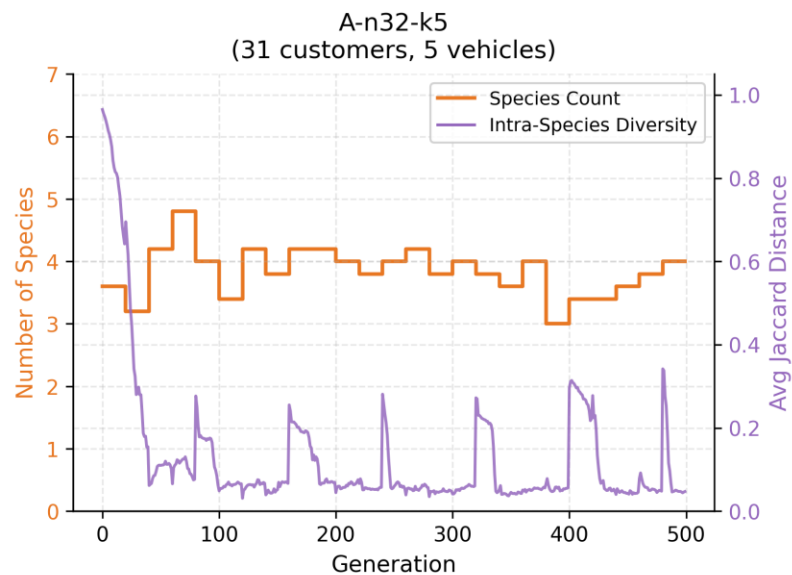
Scalability: Solution Quality



Scalability: Gap to Best Known Solution



GSA Species Dynamics (Averaged over All Runs)



GSA Species Size Evolution (Run 1)

